5.8 Vitamins

Vitamins are essential organic, compounds that are needed in small amounts in the diet both to prevent deficiency diseases and to support optimal health. The term vitamin (vital amines) was coined by Casmir Funk. The term vital denoting essential for life and amines because these compounds contained an amine functional group.

Characteristics of vitamins

- Vitamins are vital, organic, dietary substance that is necessary in only very small amounts to perform a specific metabolic function or prevent an associated deficiency disease.
- Vitamins are not synthesized by the body and therefore must be supplied through food.

Table: 7

Fat – Soluble Vitamins	Water – Soluble Vitamins
Vitamin A	Vitamin C
Vitamin D	B – Complex vitamins (8) Thiamine Riboflavin Niacin Pyridoxine Folate Cyanocobalamin Biotin Pantothenic acid
Vitamin E	
Vitamin K	

Functions of Vitamins

Each vitamin has its specific metabolic task. However the general functions are:

- Function as control agents in cell metabolism
- Components of body – tissue construction.
- Prevent specific nutritional deficiency disease, which is considered as a result of their primary role in cell metabolism.

Classification of Vitamins

Thirteen (13) recognized vitamins classified in two groups, based on their solubility in fat or in water are as follows:

Fat – soluble vitamins

The four fat – soluble vitamins – vitamins A, D, E and K are often present in the fat portion of foods, they are not easily lost from foods or destroyed by exposure to water, heat, air, or light.

Vitamin A (retinol and beta – carotene)

Vitamin A and carotene can be obtained from either animal or vegetable sources. The animal form is divided between retinol and dehydroretinol whereas the vegetable carotene can be split into four very potent groups– alpha– carotene, beta–carotene, gamma carotene and crypto– carotene.

Function

Vitamin A has a variety of functions in the body. It is required for the synthesis of rhodopsin and other light – receptor pigments in the eye and thus is essential for vision. Vitamin A is needed for normal growth and development to occur in the body, including the formation of bone and cartilage.

Healthy epithelial or skin cells, which line surfaces inside and outside the body, require vitamin A. Vitamin A plays a role in reproduction, metabolism, and immune system function also.

Digestion and absorption

Retinol esters in food are hydrolyzed by pancreatic and intestinal enzymes to form free retinol. After absorption, the retinol, is reesterified and transported to blood.

Carotenes are split in the intestines to form retinaldehyde, which is then reduced to

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